

Energy as a Virtual Token

No-Arbitrage, the Graph–Hodge Potential, and the Conserved Scalar of Cost-Accounted GSLTs

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Abstract

The cost-accounting construction on graph-structured lambda theories replaces a single, undifferentiated fuel (phlogiston) by a spectrum of signature-indexed tokens: as many “colours” of fuel as there are economically empowered actors. Reading the coloured tokens as the many manifest forms of energy in naive physics—chemical, mechanical, electrical—we ask what plays the role of energy itself, the single scalar into which the engines of physics interconvert those forms. We answer: adjoin *engines* (conversion contracts) between token types, require that no cycle of conversions manufactures value, and a *virtual token* appears—a valuation $\nu : \mathbf{Sig} \rightarrow \mathbb{R}_{>0}$ into which every colour is priced. We show that this virtual token is not an additional construct layered on top of no-arbitrage but is *definitionally identical* to it: on the graph of engines, the log-rates form a 1-cochain, no-arbitrage is exactly its closedness, and the virtual token is the potential 0-cochain whose gradient it is. The graph–Hodge decomposition then splits the conversion structure canonically into an *exact* part (the virtual token, i.e. energy) and a *cyclic* part (the arbitrage), the dimension of the latter being the first Betti number of the engine graph. “Shaking out arbitrage” is the orthogonal projection onto the exact part—the unit of a reflection, hence an idempotent monad, in pointed contrast to the non-idempotent cost monad $\mathbf{C}$. The construction compresses the $O(n^2)$ bilateral rates to the $O(n)$ potential and, because the virtual token is a *unit of account* (a monoid homomorphism out of $\mathbf{Sig}$) rather than a *medium of exchange* (an element of $\mathbf{Sig}$), it prices compound signatures without laundering authority—so the object-capability discipline survives. Finally we separate the two faces of physical energy: the virtual token is a conserved charge (a first law, exact in the lossless limit; a Lyapunov free energy with a second-law inequality once engines carry friction), but it is not yet the *generator* of the dynamics. We conjecture that closing that gap is an energy–time conjugacy with the temporal token stack, and read the ontology accordingly: the virtual token is a Noether charge, not a substance.

1 Introduction

The cost-accounted rho calculus [1] and its generalisation, the cost endofunctor $\mathbf{C}$ on the category $\mathbf{ciGSLT}$ of continued interactive graph-structured lambda theories [2], meter computation by gating each rewrite on the consumption of a matching token. Tokens are indexed by signatures $s \in \mathbf{Sig}$; the “spectral decomposition of phlogiston” [1] replaces one universal fuel by a family of signature-indexed resources, one per economically empowered actor. The companion note [2] already flirts with a physical reading of this family: signatures appear as an internal (flavour) index, token distributions as fields over the interaction surfaces, and the tropical semiring recovers cost.

This note takes the physical reading one step further, along a line suggested in recent dialogue. In secondary-school physics one meets energy in many manifest forms—chemical, mechanical, electrical,

thermal—and one meets *engines* that convert one form into another. The forms are many; energy is one. The unifying scalar is not any particular manifestation but the invariant that all manifestations share and that the engines conserve. We reproduce this move in the cost-accounted setting. The coloured tokens are the manifest forms. We adjoin engines—conversion contracts between token types, of exactly the kind already present as the exchange contract $\mathbf{Exchange}(c, v)$ of [1, App. B]. We then ask for the analogue of energy: a single scalar into which every colour converts.

Two hypotheses make the analogue exist. First, that engines are available across the token graph (convertibility). Second—and this is the load-bearing condition—that the conversion structure is *arbitrage-free*: no cycle of conversions returns more value than it consumed. Under these hypotheses a *virtual token* emerges: a common numeraire, a valuation against which all tokens are priced, and which agents holding positions across the token distribution may adopt in place of picking one existing colour as reference. Our thesis is that this virtual token is the cost-accounted transport of *energy*.

The note is organised as follows. Section 2 sets up conversion systems over the signature monoid. Section 3 proves the pivot: no-arbitrage is definitionally the existence of the virtual token. Section 4 gives the canonical decomposition of a conversion system into energy and arbitrage and identifies arbitrage-removal as an idempotent reflection. Section 5 reads the construction as compression and resolves the object-capability tension by distinguishing unit of account from medium of exchange. Sections 6–7 separate the first law (conservation, lossless) from the second (dissipation, friction). Section 8 states the energy–time conjugacy as a conjecture, and Section 9 draws the ontological conclusion.

2 Conversion systems over a signature monoid

We work over the signature structure of [2]: $(\mathbf{Sig}, *, ())$ is a commutative monoid, with $*$ the compound-signature operator and $()$ the unit. Token stacks $\mathbf{Stk}$, $S ::= () \mid s :: S$, form the free graded structure whose head cells are consumed in step with reduction [2].

Definition 2.1 (Conversion system). A *conversion system* over $\mathbf{Sig}$ is a tuple $\mathcal{K} = (T, E, r)$ where $T \subseteq \mathbf{Sig}$ is a finite set of token types; $E \subseteq T \times T$ is a symmetric set of *engines* (unordered pairs, presented with both orientations); and $r : E \rightarrow \mathbb{R}_{>0}$ is a *rate*, with the *reversibility* law

$$r(a \rightarrow b) \cdot r(b \rightarrow a) = 1.$$

An engine $a \rightarrow b$ is read operationally: one unit of the a -token is consumed and $r(a \rightarrow b)$ units of the b -token are released. We write $\Gamma_{\mathcal{K}} = (T, E)$ for the underlying (connected, without loss of generality) graph of engines.

The exchange contract of [1, App. B.9] is the paradigm engine: a persistent process $\mathbf{Exchange}(c, v)$ that swaps one c -token for one v -token and back, i.e. an engine with $r(c \rightarrow v) = 1$. Definition 2.1 allows arbitrary positive rates and, below, arbitrary finite graphs of them.

Definition 2.2 (Monoidal compatibility). $\mathcal{K}$ is *monoidal* when the token set is closed under $*$ on the fragment it names and the rates respect composition: whenever $a * a'$ and $b * b'$ are tokens and the component engines exist,

$$r(a * a' \rightarrow b * b') = r(a \rightarrow b) \cdot r(a' \rightarrow b').$$

Monoidal compatibility is the statement that the “value” of a compound authority is built from the values of its components in the same multiplicative way that $*$ builds the authority. It is the hypothesis under which the potential of Section 3 will be forced to respect $*$, and hence under which the virtual token prices multi-signatures coherently.

3 No-arbitrage is the virtual token

Fix a conversion system $\mathcal{K} = (T, E, r)$. Pass to logarithms: the *log-rate* is the edge 1-cochain

$$\ell \in C^1(\Gamma_{\mathcal{K}}; \mathbb{R}), \quad \ell(a \rightarrow b) := \log r(a \rightarrow b),$$

antisymmetric by reversibility, $\ell(b \rightarrow a) = -\ell(a \rightarrow b)$.

Definition 3.1 (Arbitrage). A *cycle* is a closed edge-path $a_0 \rightarrow a_1 \rightarrow \cdots \rightarrow a_k = a_0$. Its *yield* is $\prod_i r(a_i \rightarrow a_{i+1}) = \exp(\sum_i \ell(a_i \rightarrow a_{i+1}))$. The system exhibits *arbitrage* if some cycle has yield $\neq 1$; it is *arbitrage-free* if every cycle has yield 1, equivalently if ℓ has vanishing integral around every cycle.

Definition 3.2 (Virtual token / valuation). A *valuation* (or *virtual token*) for $\mathcal{K}$ is a function $\nu : T \rightarrow \mathbb{R}_{>0}$ with

$$r(a \rightarrow b) = \frac{\nu(a)}{\nu(b)} \quad \text{for every engine } a \rightarrow b.$$

Writing $p = \log \nu : T \rightarrow \mathbb{R}$ for the *log-price*, the condition is $\ell(a \rightarrow b) = p(a) - p(b)$; that is, $\ell = -\delta p$ is (minus) the graph gradient of a potential. When $\mathcal{K}$ is monoidal we additionally require ν to be a monoid homomorphism, $\nu(a * a') = \nu(a)\nu(a')$, equivalently p additive.

The reading of Definition 3.2 is that one unit of a and $r(a \rightarrow b)$ units of b have equal value: $\nu(a) = r(a \rightarrow b)\nu(b)$. The virtual token is a price list; the engines are value-preserving by construction against it.

Theorem 3.3 (The fundamental equivalence). *For a connected conversion system $\mathcal{K}$ the following are equivalent.*

- (i) $\mathcal{K}$ is arbitrage-free.
- (ii) ℓ is exact: $\ell = -\delta p$ for some potential $p : T \rightarrow \mathbb{R}$.
- (iii) A valuation $\nu = \exp p$ exists.

The potential p , hence the valuation ν , is unique up to an additive (resp. multiplicative) constant—a global choice of unit. If $\mathcal{K}$ is monoidal, ν may moreover be taken to be a monoid homomorphism, and is then unique up to a multiplicative character $\chi : \text{Sig} \rightarrow \mathbb{R}_{>0}$.

Proof. (i) $\Rightarrow$ (ii). Fix a basepoint $a_\star \in T$ and set $p(a_\star) = 0$. For any a , choose a path $a_\star \rightarrow \cdots \rightarrow a$ and put $p(a) := -\sum \ell$ along it. Path-independence is exactly the vanishing of ℓ around cycles (Definition 3.1), so p is well defined; by construction $\ell = -\delta p$. (ii) $\Rightarrow$ (iii) is $\nu = \exp p$, and (iii) $\Rightarrow$ (i) telescopes: any cycle has yield $\prod_i \nu(a_i)/\nu(a_{i+1}) = 1$. Uniqueness up to a constant is the connectedness of $\Gamma_{\mathcal{K}}$ (harmonic 0-cochains are constant). For the monoidal refinement, additivity of p on the generators propagates along $*$ -compatible engines (Definition 2.2); the residual freedom is a homomorphism $T \rightarrow \mathbb{R}$, i.e. a character of Sig . $\square$

Theorem 3.3 is the combinatorial core of the fundamental theorem of asset pricing [5]: a price system exists iff there is no arbitrage. What matters for us is the direction of explanation. The virtual token does not have to be *built* once arbitrage is removed; its *existence is the removal*. Energy is what remains of the conversion structure after one quotients by the arbitrage cycles.

Remark 3.4 (Gauge). The freedom in p up to an additive constant is precisely the freedom to choose where potential energy is zero; the freedom in ν up to a multiplicative constant is the freedom to choose the unit of the numeraire. Selecting an *existing* colour as reference (as one selects “prices in dollars”) fixes this gauge at a vertex of $\Gamma_{\mathcal{K}}$; the virtual token is the refusal to privilege any vertex.

4 Energy and arbitrage: the graph–Hodge decomposition

When $\mathcal{K}$ is *not* arbitrage-free, Theorem 3.3(ii) fails: ℓ is not exact. The exact obstruction is measured by discrete Hodge theory on the graph $\Gamma_{\mathcal{K}}$ [6]. Orient the edges and equip C^0, C^1 with the standard inner products. Let $\delta : C^0 \rightarrow C^1$ be the coboundary (gradient) and $\delta^* : C^1 \rightarrow C^0$ its adjoint (negative divergence). Then

$$C^1 = \underbrace{\operatorname{im} \delta}_{\text{exact (gradient) flows}} \oplus \underbrace{\operatorname{ker} \delta^*}_{\text{cycle (divergence-free) flows}},$$

an orthogonal direct sum.

Definition 4.1 (Energy and arbitrage components). Write the log-rate as $\ell = \ell_{\text{en}} + \ell_{\text{arb}}$ along this splitting: $\ell_{\text{en}} \in \operatorname{im} \delta$ is the *energy component* (an exact, hence arbitrage-free, rate structure) and $\ell_{\text{arb}} \in \operatorname{ker} \delta^*$ is the *arbitrage component* (a divergence-free circulation with no potential).

Proposition 4.2 (Arbitrage is a cohomology class). $\mathcal{K}$ is arbitrage-free iff $\ell_{\text{arb}} = 0$. The space of arbitrage components has dimension equal to the first Betti number of the engine graph,

$$\dim \operatorname{ker} \delta^* = b_1(\Gamma_{\mathcal{K}}) = |E| - |T| + 1$$

(for connected $\Gamma_{\mathcal{K}}$). In particular a tree of engines ($b_1 = 0$) admits no arbitrage, and each independent cycle contributes exactly one degree of arbitrage freedom.

Proof. The class of ℓ in $C^1 / \operatorname{im} \delta \cong H^1(\Gamma_{\mathcal{K}}; \mathbb{R})$ vanishes iff ℓ is exact iff (Theorem 3.3) $\mathcal{K}$ is arbitrage-free; and $\dim H^1 = b_1$ for a graph. $\square$

Now the operational content of “shaking the arbitrage out.” The orthogonal projection

$$\Pi : C^1 \longrightarrow \operatorname{im} \delta, \quad \Pi \ell = \ell_{\text{en}},$$

is the least-squares replacement of the observed rates by the nearest arbitrage-free rates; it recovers a potential p (hence a virtual token) up to the harmonic constant. Two properties are worth stating because they contrast sharply with the cost functor.

Proposition 4.3 (Arbitrage-removal is an idempotent reflection). Let $\mathbf{Conv}$ be the category of conversion systems on a fixed graph and $\mathbf{Conv}_0 \hookrightarrow \mathbf{Conv}$ the full subcategory of arbitrage-free ones. The map $L : \ell \mapsto \Pi \ell$ is a reflector: it is left adjoint to the inclusion, its unit $\eta_{\mathcal{K}} : \ell \Rightarrow \iota L \ell$ is the natural transformation sending each system to its arbitrage-free projection, and the induced monad ιL is idempotent ($\Pi^2 = \Pi$).

Remark 4.4 (Two monads of opposite character). The idempotence of Proposition 4.3 is the exact foil to the cost monad $(\mathbf{C}, \eta, \mu)$ of [2, §9], which is *non*-idempotent: μ merges two resource accounts rather than collapsing a redundant copy, so re-metering a metered calculus yields a finer account. Metering *accumulates*; arbitrage-removal *saturates*. One squeeze suffices to expel arbitrage, because the cyclic part of a fixed cochain is expelled once and does not return; whereas each application of $\mathbf{C}$ genuinely adds a layer of signatures and a stack-of-stacks. The two constructions are monads on the same material with opposite idempotence, and the note [2] makes both natural.

The categorical packaging of Proposition 4.3 is stated at the level of rigour a working note allows: it presumes a choice of morphisms of conversion systems (rate-nonincreasing graph maps are the natural candidate, matching the dissipative order of Section 7); pinning these down and verifying the universal property in full is deferred. The concrete content—orthogonal projection onto the exact cochains, idempotent, with η the assignment of the projected system—is exact.

5 Compression, coordination, and the unit of account

Compression

A conversion system carries, a priori, one rate per engine: up to $O(n^2)$ real numbers for $n = |T|$ tokens with all pairwise engines. Theorem 3.3 collapses this to the potential: $O(n)$ numbers, one price per token, minus the global gauge constant.

Proposition 5.1 (No-arbitrage is a rank condition). *For a complete engine graph, arbitrage-freedom is equivalent to the log-rate matrix $L_{ab} = \log r(a \rightarrow b)$ being the difference matrix of a single potential, $L_{ab} = p(a) - p(b)$; i.e. $L = p \mathbf{1}^\top - \mathbf{1} p^\top$. The bilateral data thus compress from the $\binom{n}{2}$ independent entries of a general antisymmetric L to the $n - 1$ independent entries of p .*

The compression is the coordination win read statically: agents holding positions across the token distribution need agree only on the single broadcast field $p : T \rightarrow \mathbb{R}$ rather than continuously reconcile $O(n^2)$ bilateral rates. The virtual token is the minimal sufficient statistic of a coherent conversion structure.

Unit of account, not medium of exchange

There is an immediate objection from the object-capability discipline of [1, §4.8.5]: weakening is forbidden. A compound token minted as $s_{\text{sender}} * s_{\text{compliance}}$ encodes “neither party alone may act”; an engine that converted it into a single s' would launder away the compliance requirement. If the virtual token were an engine to which *everything* converts, the multi-signature guarantee would evaporate.

The resolution turns on a distinction the physics analogy makes vivid and that Definition 3.2 already encodes: the virtual token is a *unit of account*, not a *medium of exchange*.

Proposition 5.2 (Pricing does not launder authority). *A valuation is a monoid homomorphism $\nu : (\text{Sig}, *, ()) \rightarrow (\mathbb{R}_{>0}, \cdot, 1)$, an arrow out of **Sig**; it is not an element of **Sig** and it induces no engine. It assigns $\nu(s_1 * s_2) = \nu(s_1) \nu(s_2)$ without providing any reduction that consumes an $s_1 * s_2$ -token in exchange for an s' -token. Hence pricing a compound authority is compatible with the prohibition on weakening: the number $\nu(s_1 * s_2)$ commits no holder to accepting s_1 alone.*

Nobody ever *holds* virtual tokens; one *prices* in them. A unit of account requires no custody, no issuer, and no depletable pool, and it privileges no agent’s colour—which is exactly why it lowers coordination cost below that of anointing an existing token.

Remark 5.3 (Dual to the s_0 -collapse). Setting every signature to a single distinguished s_0 recovers undifferentiated phlogiston [1, §6, Rmk. 5.1]—the monochromatic limit, which *discards* the spectrum. The virtual token is the dual move: keep the spectrum—chemical, mechanical, electrical stay distinct as colours—and adjoin a conserved scalar *above* them that makes them commensurable. Energy does not abolish the difference between chemical and electrical potential; it grades them onto one axis. The s_0 -collapse erases colour; the valuation ν measures it.

6 Conservation: the first law

Extend ν additively from tokens to stacks and multisets by $\nu(s :: S) = \nu(s) + \nu(S)$, $\nu(()) = 0$ (the temporal monoid $::$ read through the price). This is the *total value* of a token supply.

Proposition 6.1 (Value conservation in the lossless regime). *In an arbitrage-free conversion system, the total value ν of a token supply is invariant under every engine operation: an engine $a \rightarrow b$ replaces one a -cell of value $\nu(a)$ by $r(a \rightarrow b)$ b -cells of total value $r(a \rightarrow b) \nu(b) = \nu(a)$. Consequently ν is a conserved quantity of the splitter/joiner token economy of [1, App. A.6].*

Proposition 6.1 upgrades the *conservation of authority* of [1, Prop. 4.7]—the invariance of the consumed signature *multiset* across every partition of a join—to a scalar law. Where [1] conserves a multiset, the virtual token conserves its total price. This is the first law: in the reversible idealisation, value is neither created nor destroyed by conversion, only redistributed across colours.

7 Friction: the second law

The platform’s design philosophy is *valuable friction* [1, §5.3]: every resource-consuming operation carries an explicit cost. Applied to the engines themselves, this says real conversion is *lossy*. Model a lossy engine by a rate strictly below the value-preserving one: converting $a \rightarrow b$ yields $\rho(a \rightarrow b) = r(a \rightarrow b) e^{-\theta(a \rightarrow b)}$ with dissipation $\theta(a \rightarrow b) \geq 0$.

Proposition 7.1 (Dissipation breaks exactness into an inequality). *With lossy engines the log-rate cochain $\hat{\ell} = \ell - \theta$ satisfies, around every oriented cycle,*

$$\oint \hat{\ell} = - \oint \theta \leq 0,$$

with equality iff the cycle is frictionless. Hence no exact potential exists in general; instead the total value ν (computed against any fixed reference potential) is non-increasing along engine operations, and strictly decreasing whenever a lossy engine fires. ν is thus a Lyapunov function for the token economy.

Remark 7.2 (Free energy, exergy, and mortality). Proposition 7.1 is the second law arriving on schedule. The frictionless virtual token of Section 6 is a conserved *energy*; the realistic, friction-bearing virtual token is a monotone *free energy* (exergy)—the part of the value still able to do computational work, dissipated by $\oint \theta$ as an entropy analogue. This is already the ambient picture of [3], where metabolism is motion on a Ruelle–Bowen landscape and death is the absorbing set: a process whose free energy is exhausted deadlocks by starvation [3, Def. 1]. The Lyapunov ν of Proposition 7.1 is a certificate of that descent—value degrades monotonically toward the starvation boundary. The clean conserved energy is the reversible limit $\theta \equiv 0$; the cost-accounted calculus, being frictionful by design, lives in the dissipative regime, and its honest scalar is free energy.

8 The generative half: an energy–time conjecture

Physical energy wears two hats. It is the conserved charge of Sections 6–7, and it is the *generator of time evolution*—the Hamiltonian. The virtual token, so far, wears only the first hat: it is a valuation, not a generator of reduction. Yet the missing conjugate variable is already present in the construction. By [2, Prop. 4.1] the temporal token stack $::$ is the reified clock: its consumed prefix is an exact operational modulus, the number of forced redexes, the run length itself. Time, in a cost-accounted GSLT, is $::$.

Conjecture 8.1 (Energy–time conjugacy). *In an arbitrage-free, frictionless cost-accounted GSLT, the total virtual-token value ν is the conserved quantity conjugate to translation along the temporal monoid $::$, in the sense of a Noether [8] correspondence: the path-independence symmetry of*

conversion (Theorem 3.3) is the internal symmetry whose conserved charge is ν , and its coupling to $::$ -translation makes ν the generator of a step of reduction. Under friction the correspondence degrades to the second-law inequality of Proposition 7.1: ν generates a descent rather than a symmetry.

Conjecture 8.1 is offered in the spirit of the physical correspondences of [2, §13], not as a theorem. What it would buy is the second hat: an identification of the conserved charge with the generator of dynamics, which in physics is a non-trivial coincidence (the Hamiltonian is both). The cost-accounted setting is unusual in letting the two be prised apart and studied separately—the charge is manifest, the generator conjectural—and in offering a concrete candidate ($::$) for the conjugate time.

9 Ontology: a charge, not a substance

We can now answer the ontological question that motivated the note. *What is the virtual token?* It is not a substance—not a stuff one holds, transfers, or points at. Proposition 5.2 makes this precise on the syntactic side (it is an arrow out of **Sig**, never an inhabitant), and Theorem 3.3 makes it precise on the semantic side (it is the potential forced into existence by a symmetry of the conversion dynamics—the path-independence of engines). Its ontological status is that of a *Noether charge*: the conserved scalar associated with an invariance, in the deflationary reading of energy under which energy *is* “the conserved quantity associated with time-translation symmetry” rather than a substance the world contains. Both are invariants of an equivalence, not occupants of it.

Honesty requires the counter-reading. Physical energy earns a thicker ontology than mere bookkeeping because it also *generates* the dynamics: the same object that is conserved is the Hamiltonian that pushes the state forward in time, and their coincidence is substantive. The virtual token, as established here, discharges only the bookkeeping role; the generative role is Conjecture 8.1. So the sharp form of the ontological question is not settled but sharpened: *is energy a conserved charge that happens to generate time, or two coincident things?* In the cost-accounted GSLT one may hold the two apart—the charge proven, the generator open—and watch whether they want to merge. That is more than the physics affords, and it is the reason the transported notion is worth taking seriously rather than as mere analogy.

Further work

Three threads deserve promotion beyond working-note rigour. First, the reflection of Proposition 4.3: fix the morphisms of **Conv** (the dissipative order of Section 7 suggests rate-nonincreasing maps), and verify the adjunction and idempotence against them, situating the arbitrage-removal monad beside the cost monad **C** of [2] in one 2-categorical picture. Second, the free-energy reading of Section 7: identify the state function whose descent Proposition 7.1 certifies with the exergy of a Ruelle–Bowen potential on the transition graph [3], and make the “entropy production” $\oint \theta$ a genuine thermodynamic quantity rather than an analogy. Third, Conjecture 8.1: specify the symmetry action of $::$ -translation and the Noether pairing precisely enough to decide whether the conserved ν generates a reduction step, closing—or refuting—the energy–time conjugacy.

E Pluribus, Unum Pretium.

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